Project Update 4&5

Section 1. Data Collection

The XLK stock price we are looking at is accounted by S&P Dow Jones Indices and State Street Global Advisory. S&P Dow Jones Indices is in charge of creating the company benchmark on entering the technology company stock pool; while the State Street Global Advisors selected particular stocks into the combination of the XLK stock. The data we are using is collected from the investing.com website, a financial data platform that provides public real-time market data directly from stock exchanges such as the NYSE and NASDAQ. The primary purpose of collecting the data is to provide a transparent record of the performance of the Technology Selected Sector Index. The subjects of this data series are the large-cap U.S. technology companies such as Microsoft, NVIDIA, and Apple whose market valuations and returns constitute the movement of the XLK fund.

The variables we are using includes:

- Date: The specific trading day of the observed data
- Price (Close): The final price at which the ETF traded when the stock exchange closed (typically 4:00 PM EST).
- Open: The price of the very first transaction when the market opened (9:30 AM EST).
- High & Low: The maximum and minimum prices reached during the trading session.
- Vol.: The total number of shares traded during the day (e.g., 15.90M means 15.9 million shares).
- Change: The percentage difference between the current day's closing price and the previous day's closing price.

The data we are using is collected from 1/3/2017 to 3/28/2026, which incorporates the AI outbreak time. One of the limitations of this dataset is that there's a possibility for State Street to rebalance the stock composition over time, thus the data might not be consistent to offer a robustic overtime analysis.

Section 2. Variable Creation and Data Cleaning

Original dataset contains date as a character which may cause problems when using it as our time predictor, so we converted to the Date type in R. We observed outliers that are doubled the price of the remaining data points. Looking at the data, we found that for most weekends, prices are carried over from the Friday of the same week. These outliers correspond to weekend days, as indicated by their dates, and their volumes are recorded as empty strings. We decided to remove these outliers because they are not consistent with the general pattern, which they doubled the Friday price. Lastly, we split the dataset into two groups using November 30, 2022 (the release date of ChatGPT) as the cutoff. This is used for the comparison of the two time seires.

We will difference the data to stabilize variance, and remove the overall growth in the stock market as well as possible market cycles.

Section 3. Analytic Methods

EDA

Table 1: Summary Statistics of Stock Price

mean_price	sd_price	min_price	max_price
69.91683	33.95005	24.4	152.07

Table 1 shows the mean, standard deviation, minimum and maximum of the stock price between 2017/1/3 to 2026/3/27.

Transform Data

Figure 1 shows the stock price over time. The red dashed line indicates the time when ChatGPT was released. Overall, the stock price shows a general upward trend during the period shown.

Figure 2 indicates that the time series is not stationary, as the ACFs are out of the boundary and decay very slowly. Thus, we need to transform the data to make it stationary. For stock data, taking the natural log is a common way to transform the data.

Figure 3 shows the log return over time after log transformation. The log return fluctuates around zero, and the fluctuations are small for most of the time. However, there are a few

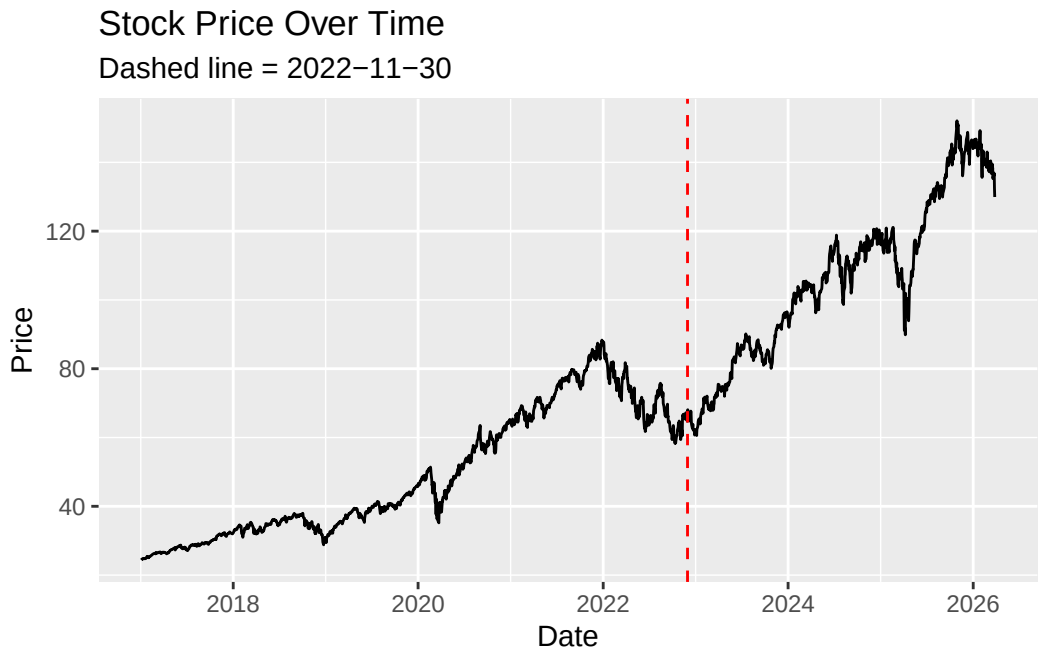


Figure 1

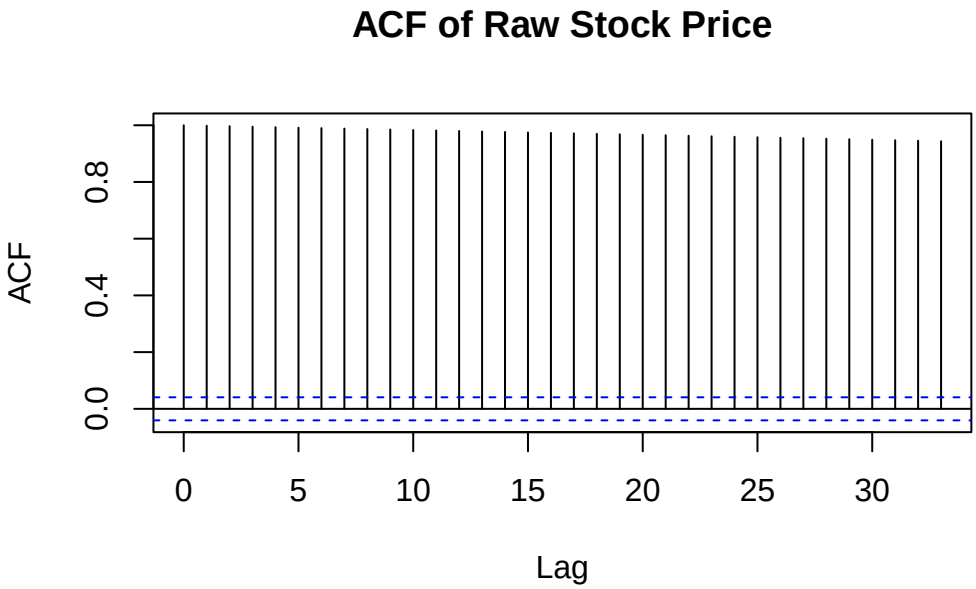


Figure 2

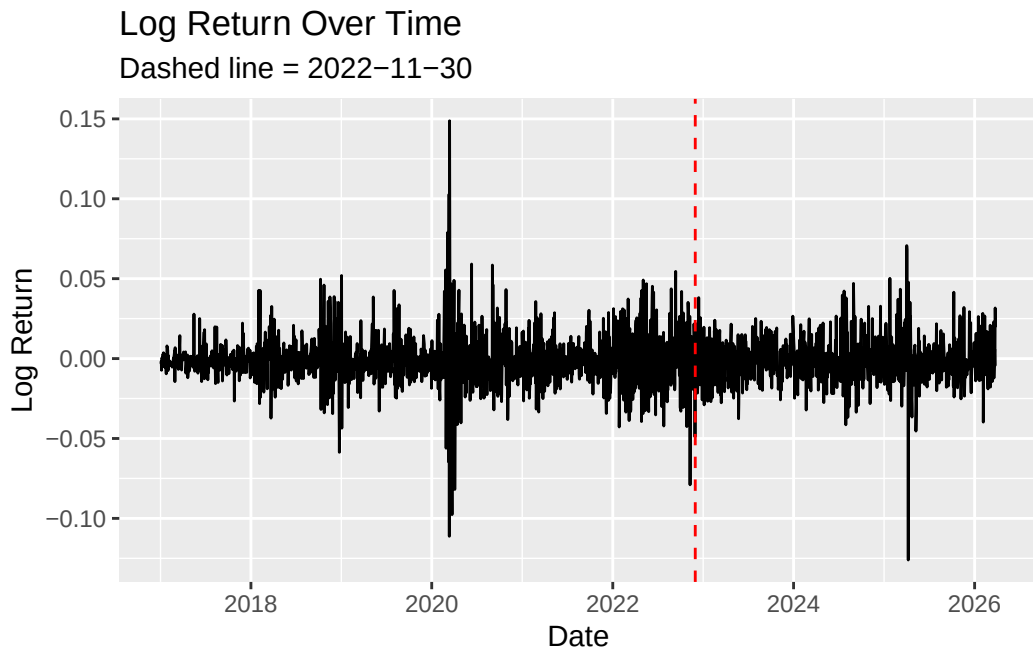


Figure 3

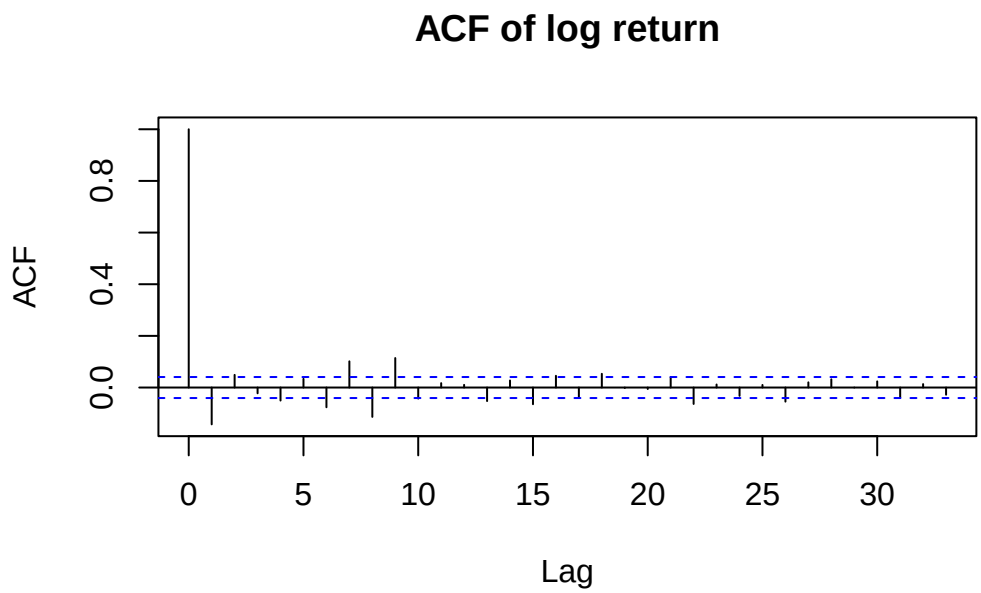


Figure 4

large spikes. The biggest fluctuations occur around 2020 when the pandemic began. This suggests big fluctuations during pandemic.

Figure 4 shows the ACF of log return. Most ACFs fall within the bound. And Figure 3 suggests the mean is around zero. Thus, the log return time series looks stationary.

Compare log return

Table 2: Summary Statistics of Stock Price

period	Mean_Return	SD_Return
After	-0.00078	0.01456
Before	-0.00069	0.01641

According to table Table 2, the mean log return before and after the appearance of ChatGPT is both close to zero.

Fit Model

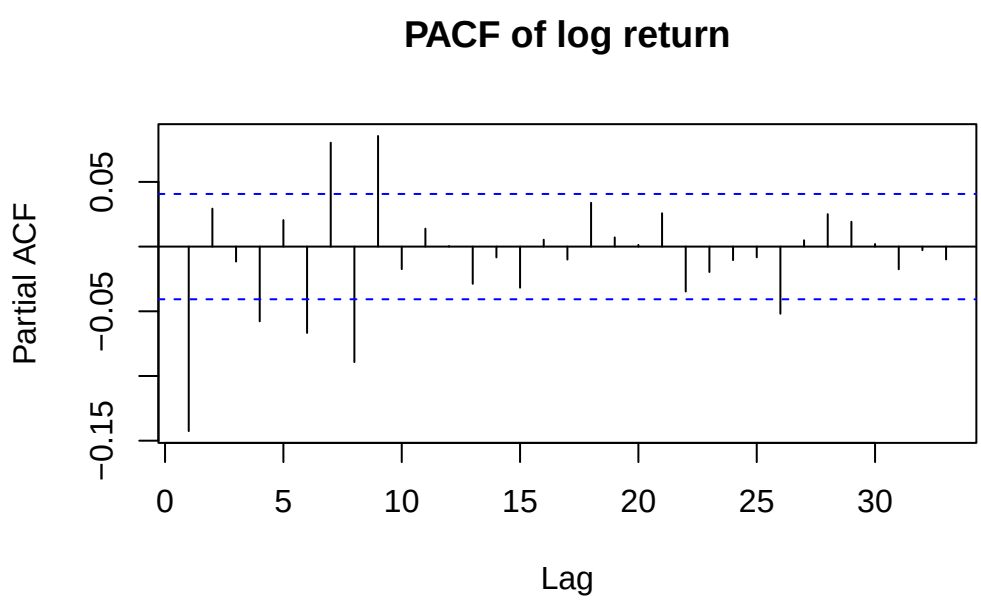


Figure 5

According to Figure 5, there is a strong pike at lag=1. So we fit AR(1), MA(1), and ARMA(1) to capture the time series.

```
Call:
arima(x = data$log_return, order = c(1, 0, 0))

Coefficients:
          ar1  intercept
      -0.1426     -7e-04
s.e.   0.0206      3e-04

sigma^2 estimated as 0.0002436:  log likelihood = 6359.22,  aic = -12712.45
```

```
Call:
arima(x = data$log_return, order = c(0, 0, 1))

Coefficients:
          ma1  intercept
      -0.1337     -7e-04
s.e.   0.0199      3e-04

sigma^2 estimated as 0.0002439:  log likelihood = 6357.63,  aic = -12709.25
```

```
Call:
arima(x = data$log_return, order = c(1, 0, 1))

Coefficients:
          ar1     ma1  intercept
      -0.3122  0.1725     -7e-04
s.e.   0.1136  0.1171      3e-04

sigma^2 estimated as 0.0002434:  log likelihood = 6360.17,  aic = -12712.34
```

Model Comparison

Table 3: Comparison of AR(1), MA(1), and ARMA(1,1) Models

Model	Coefficients	AIC
AR(1)	ar1 = -0.1426	-12712.45
MA(1)	ma1 = -0.1337	-12709.25
ARMA(1,1)	ar1 = -0.3122, ma1 = 0.1725	-12712.34

AR(1), MA(1) and ARMA (1,1) are fitted to the log return series. The AR (1) has the smallest AIC among all the three models. The ARMA (1,1) has a slightly higher AIC compares to AR(1), while the MA(1) has the overall largest AIC.

Forcasting

We are interested in how significant the data after 2022/11/30, when AI stared its prevalence, is to forecasting future stock market prices. We will being using data frames of different time sets: 2017/1/3 - 2026/3/27, 2017/1/3 - 2022/11/30 and 2022/11/30 - 2026/3/27. The forecast models we will be using are five "benchmark" methods: Mean method, Naive method and Seasonal naive method. Drift method and decomposition method seem unnecessary with log return data because we already removed the trend. (Note: We can do a regression forecast if we have time)

First, we aimed to examine whether AI has had an unexpected impact on the stock market and contributed to its fluctuations. We would like to use data from 2017/1/3 - 2022/11/30 to forecast data after 2022/11/30.

Table 4: Forecast Errors

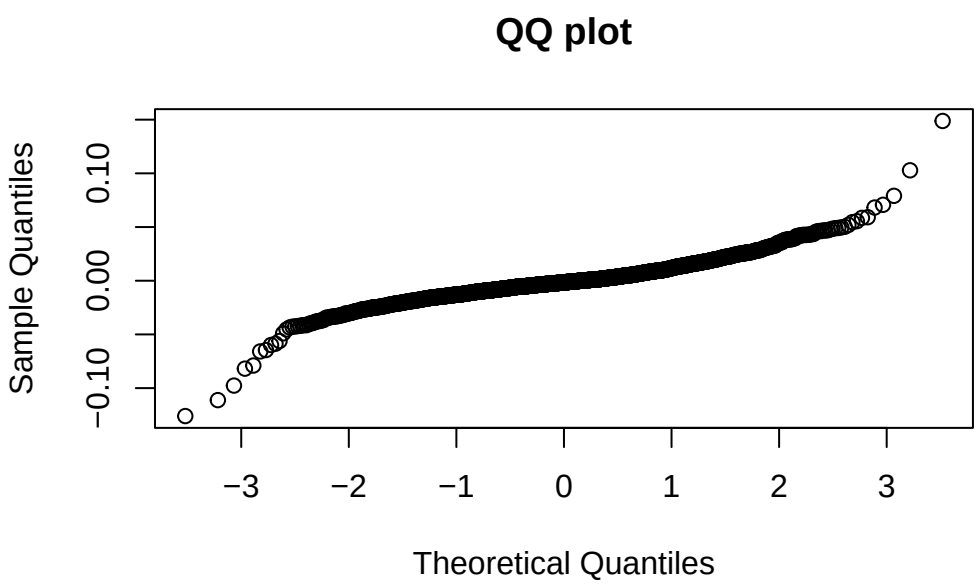
Method	MAE	RMSE
Mean	0.01049	0.01457
Naive	0.01241	0.01597
Seasonal	0.02161	0.02880

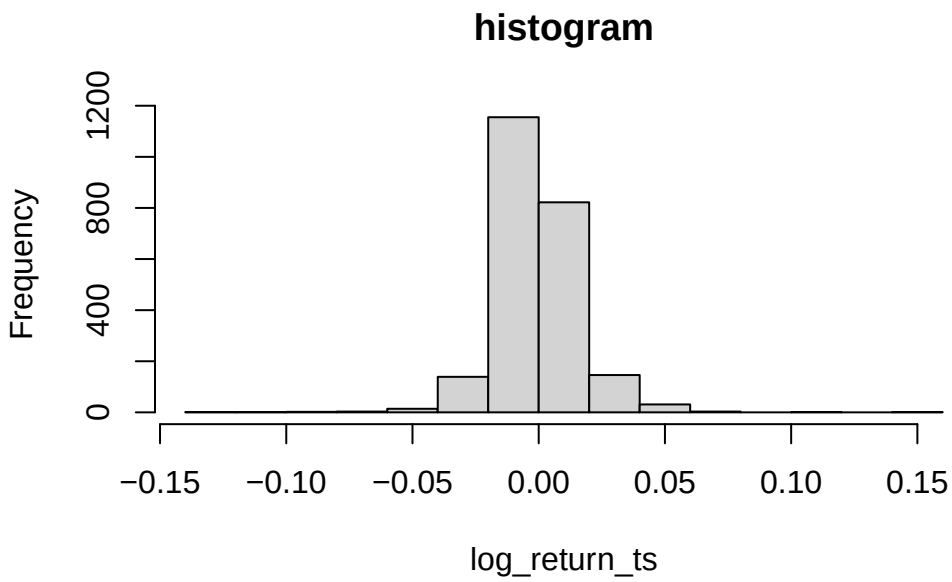
According to Table 4, all the scaled dependent errors are small, even for these easier approaches. It appears that the data prior to 2022/11/30 can effectively predict the data following the cutoff point. Furthermore, this suggests that AI have not yet contribute a significant impact in the stock market. As a result, we will use the entire time frame, allocating 80% for training and 20% for testing, to build a forecast model in predict prices in 2026.

Table 5: Forecast Errors-full model

Method	MAE	RMSE
Mean	0.01128	0.01631
Naive	0.01459	0.01845
Seasonal	0.01311	0.01754

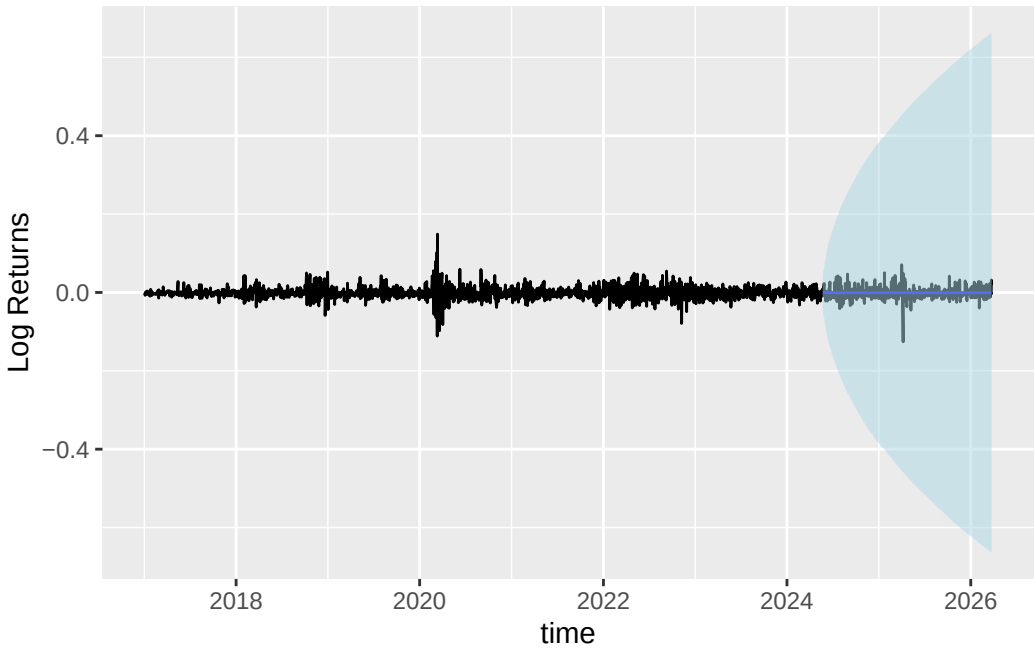
Table 5 shows the mean method is the best with lowest MAE and RMSE. We did not put MAPE here for comparison, because our log return data moves around 0, where some small values here make MAPE enormous and not as a good parameter to look at.





Next, we would like to also make a prediction interval for this forecast model. Our ACF for log return looks acceptable as previously showed. The histogram looks normal, however qq-plot reveal that log return does not satisfy the linearity assumption. This suggests that our prediction interval is off.

(Note: actually I just realize there is probably no need to do prediction on this, because it is around 0; we are wondering if we can forecast in ARMA)



Summary

The project test whether the rise of AI has significantly impacted the behavior of technology sector stock returns. Using the XLK ETF price data from 2019 to 2026, we transformed the series into log returns to stationaize the data and applied AR(1), MA(1), and ARMA (1,1) time series models. The result suggests that stock return pattern remained relatively stable befoe and after the emergence of AI, indicating that AI has not yet had a significant impact on the technology stock market price. More broadly, this shows that while AI is often seen as a transformative force, its impact on financial markets may not be immediately significant on stock market return fluctuation.

Limitation

There are some potential limitations for the dataset. One of the limitations is that there's possibility for State Street to rebalance the stock composition over time, thus the data might not be consistent to offer a robust overtime analysis.

Another limitation is the dependency of the dataset. In the EDA, we found out the time series data points are highly correlated whether before or after the data transformation. For instance, the QQ plot indicates deviations from normality, suggesting that model assumptions may not

hold and prediction intervals may be unreliable. In addition, our model only test on the short-term impacts, but AI boom may results from a long-term trend. The log transformation may mask the underlying long term trend.

Also, AR, MA, ARMA are all autoregressive models that look solely on the past values. They do not include other important factors, such as policy changes, economic policy changes, etc. There are some other factors that are not included in our models .